

Preliminary Tasks

As a developer,

I want the architecture and workflow to be well defined
So we can collaborate and split work up easily.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Make template for architecture for game / main loop for pygame	Henry Michael Hoopes	1 hour(s)	Sept 3rd, 3:30 p.m. - 4:30 p.m.
Split template up into separate files	Henry Michael Hoopes	.1 hour(s)	Sept 3rd. 6:20 p.m.
Add button class/ template	Henry Michael Hoopes	1 hour(s)	Sept 8th, 8:00 a.m. - 9:00 a.m.
Add template functionality for each screens	Henry Michael Hoopes	.75 hour(s)	Sept 8th, 9:45 a.m. - 10:30 a.m.
Split into branches	Henry Michael Hoopes	.1 hour(s)	Sept 8th, 10:30 a.m.
Integrate all work	Henry Michael Hoopes	1.5 hour(s)	Sept 13th, 7:30 a.m. - 9 a.m.

Bugfix Tasks

As a player,

I want no unintentional / unexpected functionality
So that I can experience the game in the desired manner.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Patch for filling screen with flags to win	Henry Michael Hoopes	.25 hour(s)	Sept 17th, 4:15 p.m. - 4:30 p.m.

Added row/col numbers	Henry Michael Hoopes	.25 hour(s)	Sept 17th, 4:55 p.m. - 5:10 p.m.
Added board to win/lose screen	Henry Michael Hoopes	.5 hour(s)	Sept 17th, 2:00 - 2:30 p.m.

User Story 1 — Sprite Sourcing

As a user

I want clear symbols for flags, mines, and indicators
So that I can better visually understand the game.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Make symbols for flags	Ryan Grimsley	.75 hours	.5 hours, Sep 3: 6:30pm-7pm
Make symbols for mines	Ryan Grimsley	.75 hours	.5 hours, Sep 3: 7pm-7:30pm
Make symbols for indicators	Ryan Grimsley	.75 hours	.5 hours, Sep 3: 7:30pm

User Story 2 — Start Screen

As a user

I want a start screen when I start the game
So that I can input the number of mines and have an entrypoint to the game.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Design Start Screen	Ryan Grimsley	.5 hours	1.5 hour, Sep 8: 11am-11:30am and 1pm-2pm
Upload to Github	Ryan Grimsley	.1 hours	.1 hours, Sep 8: First commit for story at 6:30pm

			Sep 15: second commit for story at 5:30pm
Implement start screen in UI Engine	Ryan Grimsley	1 hour	2 hours, Sep 8: 2pm-2:30pm and 5pm to 6:30pm
Add input box for number of mines (use arrow keys to increase or decrease)	Ryan Grimsley	.5 hours	1 hour, Sep 15: 1pm-2pm
Add event handling for starting game with number of mines	Ryan Grimsley	.5 hours	1 hours, Sep 15: 2pm-2:30pm, 5pm-5:30pm

User Story 3 — Right Click

As a player,

I want a right click to place or remove a flag,
So that I can mark a suspected mine.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Add event handling for right click	Henry Michael Hoopes	.25 hour(s)	Sept 8th, 9:30 a.m. - 9:45 a.m.
Adjusted event handling for right click with board offset	Connor Williamson	.25 hours	Sept 21st 5:30pm - 5:45pm

User Story 4 — Left Click

As a player,

I want a left click to reveal if a cell is a mine or not,
So that I can progress the game.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Add event handling for left click	Henry Michael Hoopes	.25 hour(s)	Sept 8th, 9:15 a.m. - 9:30 a.m.
Adjusted event handling for left click with board offset	Connor Williamson	.25 hours	Sept 21st 5:30pm - 5:45pm

User Story 5 — Win Screen

As a player,

I want a screen to tell me if I won

So I know if I won.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Design layout for win screen	Henry Michael Hoopes	.25 hour(s)	Sept 8th, 10:25 a.m - 10:40 a.m.
Implement win screen in UI Engine	Henry Michael Hoopes	.5 hour(s)	Sept 8th, 10:40 a.m - 11:10 a.m.

User Story 6 — Game Engine

As a developer,

I want a robust and efficient game engine

So that the game runs smoothly and is easily maintainable.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Design logic for creating the game	Benjamin Stonestreet	1 Hr	1 Hr Sep 12: 5Pm - 6Pm

board			
Add function for decrementing and incrementing mines	Benjamin Stonestreet	15 Mins	15 Mins sep 12: 6:00 Pm-6:15 Pm
Add logic for revealing board spaces	Benjamin Stonestreet	1 Hr	1 Hr Sep 12: 6:15 - 7:15
Implemented check win function	Connor Williamson	.25 hours	.25 hours Sept 15 5:00pm - 5:15pm

User Story 7 — Board Design

As a developer,

I want a 10x10 board that allows the user to interact with the board by revealing squares, placing flags, showing how many bombs border a square and the amount of time spent on the game so far,

So I can know my progress through the game.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Created 10x10 board	Mya Hoersdig	1 hour	Sept 15 8-9pm
Added logic to display the amount of bombs a revealed square is touching	Mya Hoersdig	30 mins	Sept 15 9-9:30pm
Added logic to see if a revealed square is a mine	Mya Hoersdig	30 mins	Sept 15 9:30-10pm
Added restart button on playing screen	Mya Hoersdig	30 mins	Sept 15 10-10:30pm

Added framework for a timer to appear on the playing screen	Mya Hoersdig	30 mins	Sept 15 10:30-11pm
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User Story 9 — Documentation

As a developer,

I want clear and robust documentation for the project

So that the project is more easily understandable for any developers working on it.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Add sufficient documentation to each file/code	Benjamin Stonestreet	1 hour	1 hour, Sep 20: 10-11PM
Make diagrams for project architecture	Ryan Grimsley	1 hour	1 hour, Sep 17: 6pm-7pm
Make summary of classes and key components	Ryan Grimsley	1 hour	1.25 hours, Sep 17: 8:30pm-9:45pm

User Story 10 — Lose Screen

As a player,

I want to have a screen tell me that I lost,

So I can know if I lost.

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Design layout for lose screen	Abdelrahman Zeidan	1	.75 Sep 7 th : 12pm-2pm. Sep 8 th 2pm-3pm

Implement Lose Screen in UI Engine	Abdelrahman Zeidan	.75	.5 Sep 9 th :11am-11:30am
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User Story 11 — Emoji

As a player

I want an emoji indicator to show game state
So that I can know the games state

Task Breakdown:

Tasks	Completed By	Estimated Person Hours	Actual Person Hours / Dates
Add emoji sprites for playing screen	Abdelrahman Zeidan	.75	.5 hours Sep 14 th :1pm- 1:30pm
Add emoji sprites for lose screen	Abdelrahman Zeidan	.5	.23 hours Sep 15 th : 2pm-2:15pm
Add emoji sprites for win screen	Abdelrahman Zeidan	.5	.25 Sep 15 th : 2pm-2:15pm